Result BattleBot BB-08

Discussion of Intended Result vs. Achieved

ABSTRACT

This document evaluates the outcomes of the BattleBot project against the original objectives set at the start of Period 3. The robot was required to complete a series of weekly assignments and ultimately perform a full track run involving cone retrieval, line following, and obstacle avoidance. This discussion compares what was planned with what was actually achieved, and reflects on the factors that influenced the results.

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Battle bot

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Discussion of Intended Result vs. Achieved Result BattleBot BB-08

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1. Introduction

This document evaluates the outcomes of the BattleBot project against the original objectives set at the start of Period 3. The robot was required to complete a series of weekly assignments and ultimately perform a full track run involving cone retrieval, line following, and obstacle avoidance. This discussion compares what was planned with what was actually achieved, and reflects on the factors that influenced the results.

2. Intended Results

Primary Objective: Build a robot capable of autonomously completing a full track race picking up a cone, following a black line, avoiding obstacles, and stopping at the finish line without any manual input.

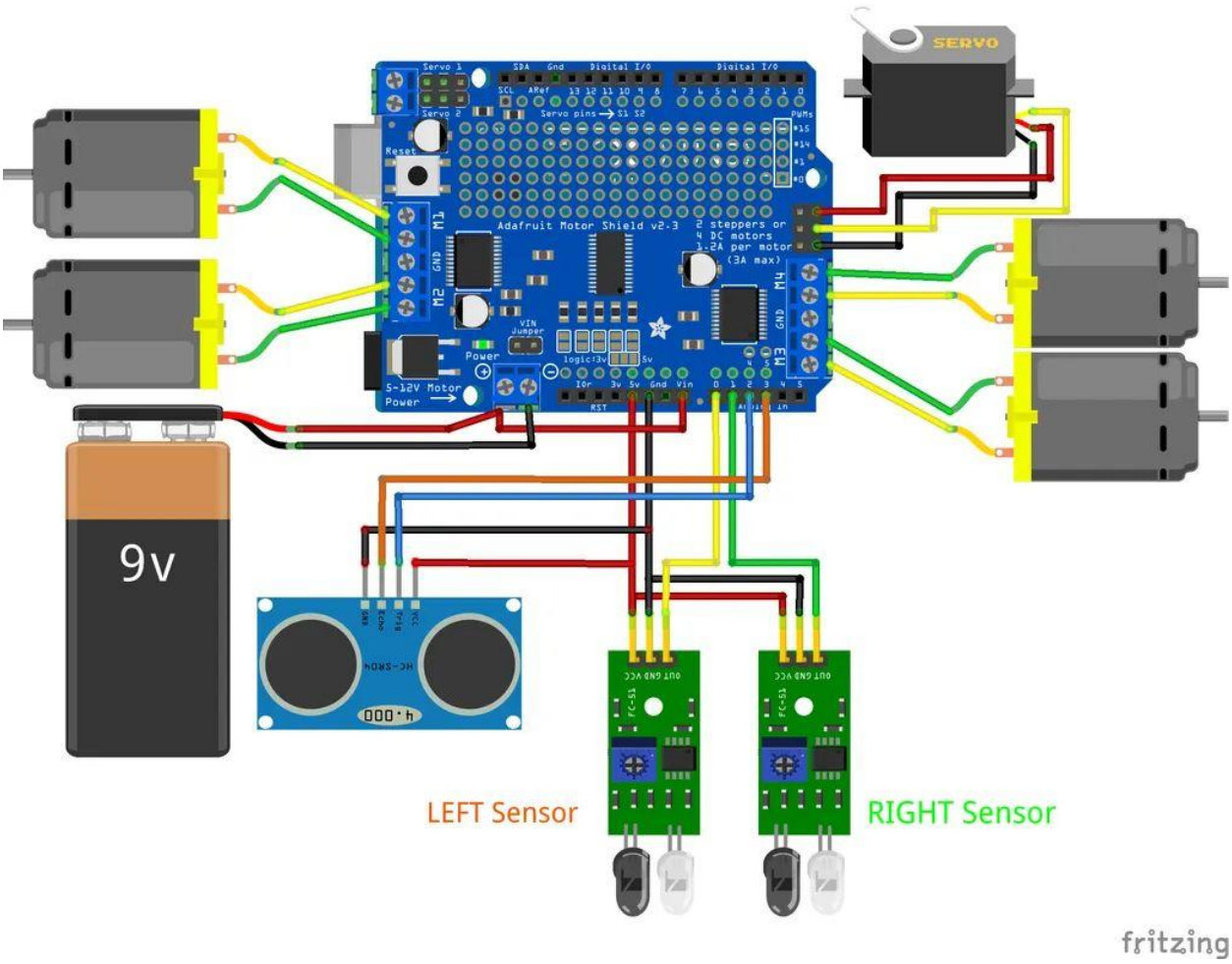
Secondary Objectives:

- Implement all individual weekly assignments: Blink, Traffic Lights, Basic Moves, Object Avoidance, Stay on Track, Grab-IT, and Game On
- Use millis() for non-blocking servo control in the Grab-IT assignment
- Use arrays for sensor reading in the line following assignment
- Integrate all functions into one combined program using a state machine

Performance Targets:

- Robot completes the full track without stopping or going off course
- Gripper successfully picks up and holds the cone throughout the run

- Obstacle is detected and avoided, with the robot returning to the original line
- Robot stops automatically when detecting the finish line (all 8 sensors on black)



3. Achieved Results

Primary Outcome: The robot successfully completed the full track run. It waited for the flag to be removed, opened the gripper, drove forward to the cone, grabbed it, turned onto the line, followed the track, avoided an obstacle using a timed bypass routine, and stopped at the finish area. After stopping, it released the cone and reversed.

Secondary Outcomes:

- All weekly assignments were completed individually and then combined into the final Game On program

- millis() was used throughout to keep the servo active without blocking other functions
- An 8-sensor array with weighted position values was used for smooth line following
- A full state machine (enum RobotState) was implemented to manage all phases of the run

Performance:

- The robot completed the track successfully in testing
- Cone was picked up and held during the full run
- Obstacle avoidance worked using a timed right-left-left-right bypass sequence
- The finish line was detected when all 8 sensors read black for 180ms, triggering the stop state

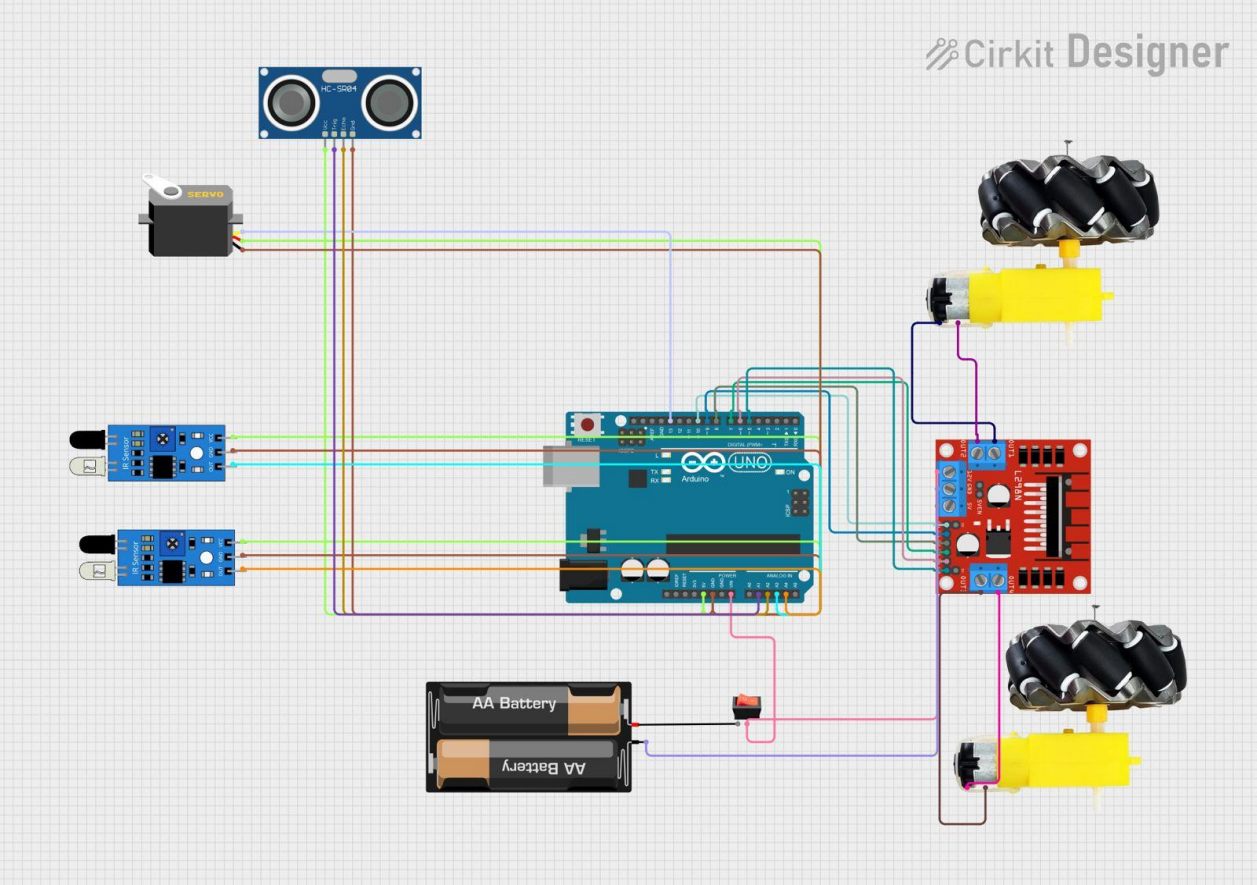
4. Comparison and Analysis

Objective	Intended	Achieved	Notes
Full autonomous track run	Complete without manual input	Achieved	Robot ran full track independently
Cone grab with millis()	Servo active without blocking	Achieved	updateServo() runs every loop cycle
Line following with arrays	Smooth tracking using sensors	Achieved	Weighted average of 8 sensors used
Obstacle avoidance	Detect and bypass, return to line	Achieved	Timed bypass with 800ms ignore window after
Finish line detection	Stop when all sensors detect black	Achieved	180ms threshold prevents false stops
Motor straightness	Drive in straight line	Partially	Left speed 224, right 200 needed to compensate for motor difference

Analysis: The primary objective was fully met. The biggest challenge during development was getting the robot to drive in a straight line the two motors ran at different speeds so the values had to be manually tuned (LEFT_SPEED 224, RIGHT_SPEED 200). The obstacle avoidance timing also required multiple test runs to get the turn angles and forward distances right. The line following logic worked well using the weighted sensor average, which allowed the robot to smoothly correct its position rather than making sharp jerky movements.

5. Factors Affecting Results

- **Motor inconsistency:** The left and right motors did not run at equal speed, requiring manual calibration of speed values
- **Timing-based movement:** Since the robot uses timed turns instead of encoders, small surface variations or battery level changes could affect accuracy
- **Sensor sensitivity:** The black threshold of 800 had to be tuned to the specific lighting conditions of the track environment
- **Obstacle avoidance window:** An 800ms ignore period after avoiding was needed to prevent the robot from immediately re-triggering avoidance when returning to the line



6. Conclusion

The BattleBot project achieved all of its intended objectives. The robot successfully completed the full track race autonomously, including cone retrieval, line following, obstacle avoidance, and automatic stopping. The main deviations from the original plan were minor calibration issues with motor speed and timing, which were resolved through iterative testing. The project demonstrated that a complete autonomous system can be built by combining individual weekly assignments into a structured state machine, and provided valuable hands on experience with embedded programming, sensor integration, and hardware-software interaction.